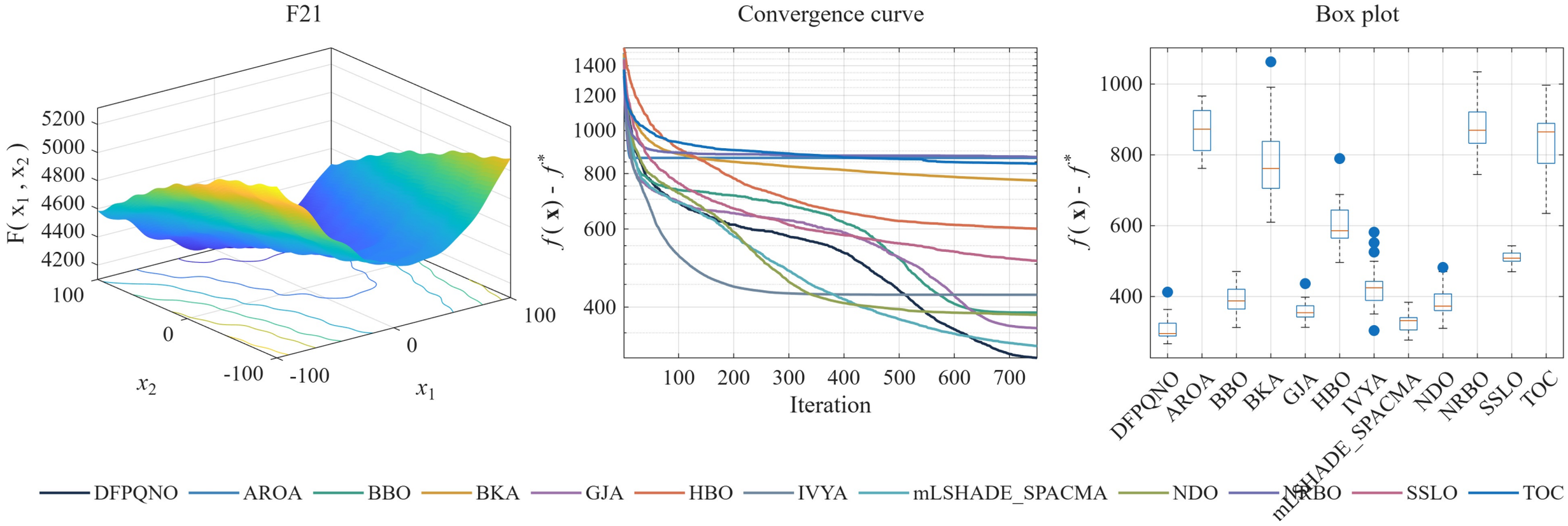

F21
F(x1 , x2)
5200
5000
4800
4600
4400
4200
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
1400
1200
1000
800
600
400
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
1000
800
600
400
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC